Propensity Scores

By Jeff Fine

What is a Propensity Score

- Defined as “The propensity score is the conditional probability of assignment to a particular treatment given a vector of observed covariates.”
- A statistical method to account for confounding bias between groups of participants in an observational study by mimicking randomized design.
- 4 Different types: Matching, Stratification, Regression Adjustment, and Weighting

Matching and Stratification

- Propensity score matching involves finding a control subject with a propensity score that is equal, or nearly equal, to the propensity score for each treatment subject

- Stratification involves dividing the entire sample into a set number of strata based on rank-ordered propensity scores and analyses are performed within each stratum

Regression Adjustment and Weighting

- Regression adjustment refers to including the propensity scores as a covariate in the regression model that is used to estimate treatment effect

- Weighting observations based on the inverse of the estimated propensity scores (commonly known as IPW).

Propensity Scores

- Just like multivariable regression, Propensity scoring requires complete data from a case in order for it to be included in the analysis.
  - Impute missing data when appropriate
- Propensity Scores are typically used for when you have a binary treatment variable.

Propensity Score: Variable Selection

- The goal of the Propensity Score methods is to achieve balance in observed confounders

- Guideline:
  - A propensity model should include only variables that affect an outcome or outcome and exposure (confounders)

Propensity Score: Assessing Model Adequacy

- The success of propensity-score modelling can be evaluated by comparing the balance between treated and control participants after accounting for the propensity score
- A common approach, in the context of propensity-score matching, is the use of statistical significance testing to compare means or proportions between the matched groups
- Compute the standardized difference of each covariate

Propensity Scores vs. Regression Model

- Simpler to determine whether the propensity score model has been adequately specified than to assess whether the regression model relating treatment assignment and baseline covariates to the outcome has been correctly specified
- When using propensity score matching or weighting, the propensity score can be estimated, and a matched or weighted sample can be constructed without any reference to the outcome.
- Propensity Score allows to ‘adjust’ for more covariates when the outcome is rare and treatment is common

Most used is Matching (PSM) and Weighting (IPW)

- PSM: creates a quasi-randomization to reduce confounding biases using either “greedy” or “optimal” matching
  - Greedy: Assigns control case nearest to treatment case based on propensity score.
  - Optimal: Matches controls to cases based on minimizing the total absolute distance between control propensity scores and treatment propensity scores
  - Something to consider is Caliper Distance
- Weighting: creates weights based on the inverse of the propensity scores and if the person is in the treatment or non-treatment group.

Table 1. Comparison of multivariable regression, propensity score matching, and inverse probability weighting by the underlying assumptions

Analytic Approach	Features			
	Causal Estimand	Identifiability Assumptions	Residual Confounding/Positivity Violations	Model Specifications
Multivariable Regression	• Conditional effects within the covariate strata • Marginal effect assuming no effect measure modification by any of the measured covariates or via standardization	• Conditional exchangeability based on the covariates used in the outcome model • Consistency for the exposure of interest • Positivity conditional on the covariates	• Positivity violation if only exposed or unexposed individuals are present within the covariate strata	• Outcome model conditional on an exposure and measured covariates
PS Matching	• Marginal effect in the population represented by the matched sample, which excludes individuals with extreme PS values from the original sample.	• Conditional exchangeability based on the covariates used in the PS estimation • Consistency for the exposure of interest • Positivity conditional on the PS.	• Potential residual confounding due to wide caliper distance • Positivity is ensured by excluding unmatched individuals.	• Propensity model conditional on measured covariates. • Outcome model can be used after matching with caution.
IPW	• Conditional effects by including an additional covariate in the weighted outcome model • Marginal effect in the target population	• Conditional exchangeability based on the covariates used in the PS estimation • Consistency for the exposure of interest • Positivity conditional on the PS	• Potential positivity violation is detected as extremely large or small weights, which can be discarded before weighting.	• Propensity model conditional on measured covariates. • IP-weighted outcome model conditional on exposure (and an additional covariate if estimating conditional effects).

IPW, inverse probability weighting; PS, propensity score.

Methods	Advantages	Disadvantages
Matching	Superior at reducing bias compared with stratification and covariate adjustment Presentation of analyses and results is very similar to randomized trial* Transparent†	Unmatched treated participants, and possibly unmatched control participants, are excluded from the analysis, decreasing precision of the treatment effect estimate and external validity Requires more control participants than treated participants
Stratification	Uses all data	Reduces bias less than matching and weighting Does not work well with survival analysis
Covariate adjustment	Uses all data	No clear distinction between the design phase and the analysis phase Assessing balance in baseline covariates between treatment groups is more cumbersome compared with the other methods Produces only odds and hazard ratios and leads to biased estimates of these ratios Requires the specification of a regression model for the relationship between the outcome and the propensity score Outcome always in sight, so temptations toward an anticipated model is always present
Inverse probability of treatment weighting	Superior at reducing bias compared with stratification and covariate adjustment Presentation of analyses and results is very similar to randomized trial* Uses all data	May be more sensitive to mis-specification of the propensity-score model and extreme propensity-score values

Method	Objectives	Details	Pros	Cons
Matching	Estimate ATT where the treatment group only includes the patient samples for which their matches are found in the original data set.[17,20]	The estimated effect is for matched samples as unmatched samples are excluded.	Popular method; it is easy to implement. Extrapolation is likely to be avoided by using PS. Conditionally exchangeable.	Bias may be introduced depending on the choice of caliper.
Stratification	Estimate the treatment effect within strata of the propensity score via ATE or ATT.[20] A summary estimate can be also obtained.[17]	Whether ATE or ATT is estimated depends on the specification of weights.[20] An additional method can be combined to obtain a summary estimate.[17,65]	Low risk of excluding samples as matching is constructed by strata rather than individuals.	Bias may be introduced depending on the choice of caliper.
Covariate modeling	Estimate the PS conditional treatment effect.[17]	The treatment effect is adjusted for propensity scores that were estimated using baseline covariates in a logistic regression model. The effect is neither ATT or ATE.[20]	Intuitively clear solution to balance the measured covariates/confounders at baseline for estimating the treatment effect.	Models for both PS and outcome need to be correctly specified (called *doubly unrobust*), which is hard to confirm.
IPTW	Estimate either ATE[11] or ATT.	Whether ATE or ATT is estimated depends on the specification of treatment effect.[20]	ATE and ATT can be estimated. It is reported that the method had yielded a less biased estimate of hazard ratios compared to stratification and covariate modeling via simulations.	In case of ATE, the estimated effect may differ greatly from the estimated results using other methods, which will require careful evaluation in light of the method objective.[66] The performance to estimate ATE may be poor if the weights for a few subjects are very large. In such a case, more restrictive assumptions are required.
Doubly robust methods	Estimate the marginal treatment effect[29–31] (ATE).	The treatment effect is estimated using doubly robust estimator which consists of PS-weighted observed outcome and 2 bias terms using observed and predicted outcome adjusted for baseline covariates. PS-matched samples are used.	Only one of the two models, for propensity score and for outcome, needs to be correctly specified (*doubly robust*). Methods for matched samples provide better estimates compared to unmatched samples.[20,42]	There are not many applications in literature to assess the method performance relative to other methods.
Multivariable regression (non–PS-based approach)	Estimate the treatment effect adjusting for the covariates in the model.	The relationships between outcome and covariates are directly specified in the multivariable regression model. The model specification may be influenced by the hypothesis of experimentalists and may be a source of bias.[20] The effect is neither ATT nor ATE.[20]	Conventional method in epidemiological studies compared to PS methods.	When the number of observations per covariate is limited (generally <8-10), there is an overfitting risk. Model specification may be subjective and the assumption of linearity of the model is restrictive.

Post-Propensity Score Analysis

- Once Propensity Score Matching is completed, a comparison of outcomes between the treated and control groups can be estimated using statistics that account for **matched** design

- Once Weights are computed for each participant, these will be incorporated in analyses when comparing outcomes between treatment and control groups.

Example using SAS

- Pharmacy Resident project looking at Burn ICU population
- Wanted to compare fluids/kg/tbsa for those who received a specific sedative (DEX) and those who did not receive DEX in the first 24 hours.
- Was concerned regarding smallish sample size (n=55 in DEX group and n=115 in non-DEX group)
- Wanted to originally do a propensity score matching on TBSA, Age, and Inhalation Injury.

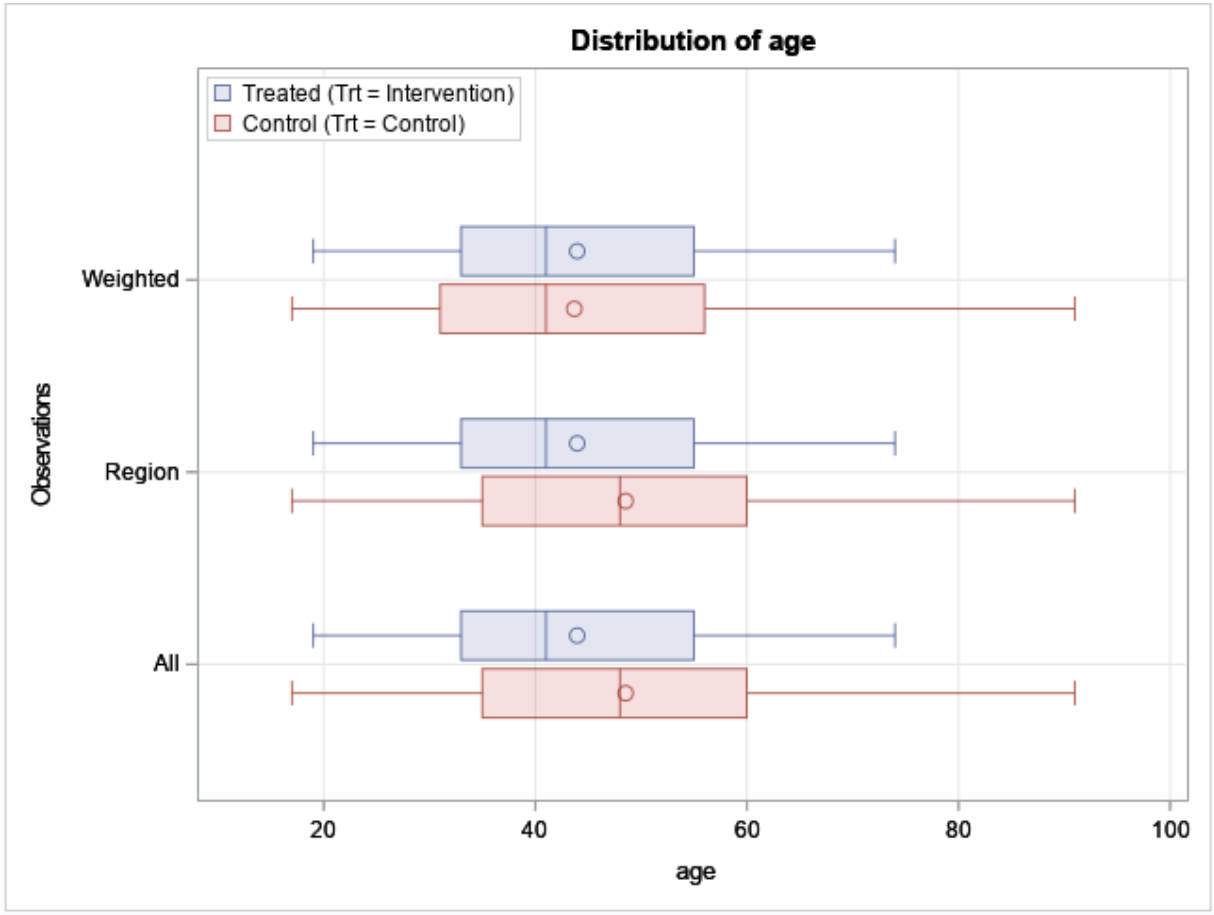
Example using SAS

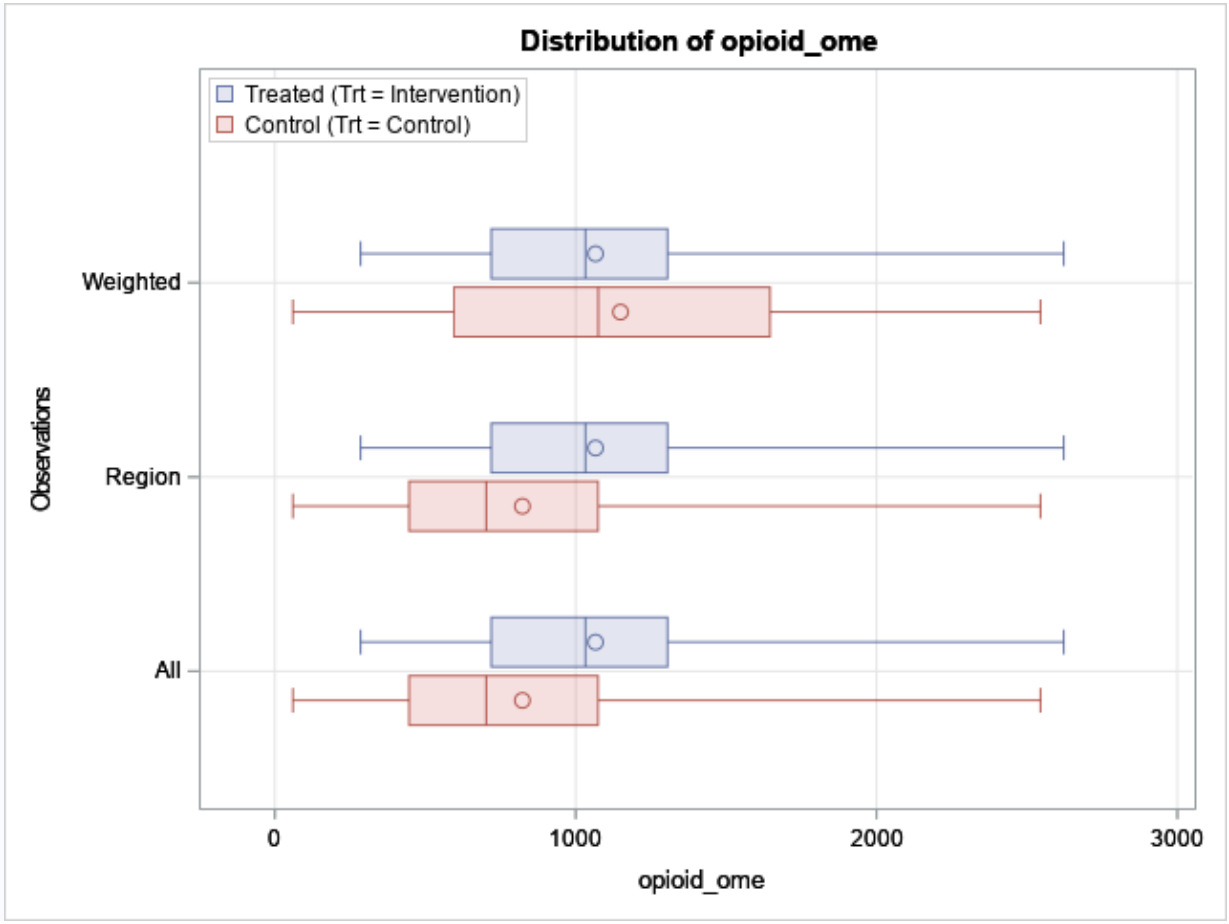
- Variables for the propensity score model was selected from what was significant (or close to significant) in the univariate analyses when comparing the two treatment groups. Significant variables included:
  - age (p-value = 0.0627)
  - Part and Full TBSA (p-value = 0.0910, p-value = 0.0672)
  - Inhalation Injury (p-value = 0.0549)
  - Opioid (OME) (p-value = 0.003)
  - Continuous opioids over 24 hours (p-value = 0.0005)
  - Continuous sedatives over 24 hours (p-value = 0.0725)

Standardized Differences

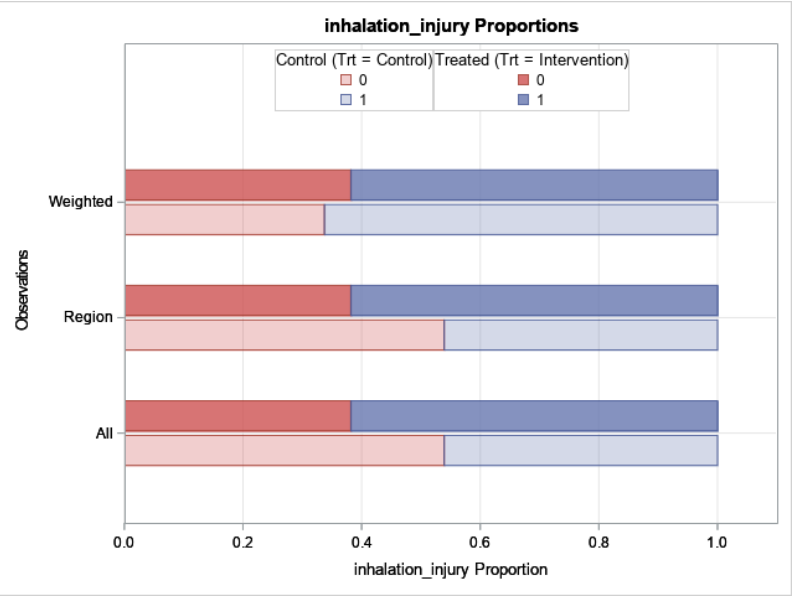
Standardized Mean Differences (Treated - Control)						
Variable	Observations	Mean Difference	Standard Deviation	Standardized Difference	Percent Reduction	Variance Ratio
Logit Prop Score	All	0.4606	0.6563	0.70186		1.0321
	Region	0.4606		0.70186	0.00	1.0321
	Weighted	-0.0796		-0.12131	82.72	0.6840
age	All	-4.5755	15.3235	-0.29859		0.7103
	Region	-4.5755		-0.29859	0.00	0.7103
	Weighted	0.2671		0.01743	94.16	0.6681
opioid_ome	All	243.1009	499.0361	0.48714		0.7884
	Region	243.1009		0.48714	0.00	0.7884
	Weighted	-82.9714		-0.16626	65.87	0.5856
inhalation_injury	All	-0.1573	0.4922	-0.31962		0.9500
	Region	-0.1573		-0.31962	0.00	0.9500
	Weighted	0.0445		0.09051	71.68	1.0560
opioid_continuous	All	-0.2435	0.4313	-0.56453		1.8179
	Region	-0.2435		-0.56453	0.00	1.8179
	Weighted	0.0278		0.06454	88.57	0.9804
sedative_continuous	All	-0.0949	0.3378	-0.28086		1.8737
	Region	-0.0949		-0.28086	0.00	1.8737
	Weighted	0.0289		0.08551	69.55	0.8945
Standard deviation of All observations used to compute standardized differences						

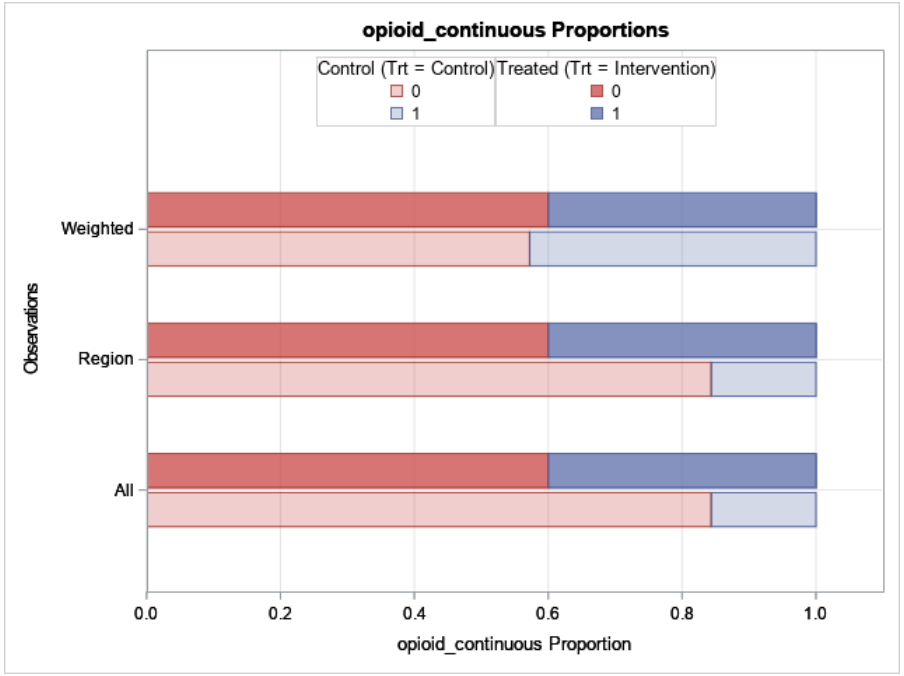
Distributions after weighting

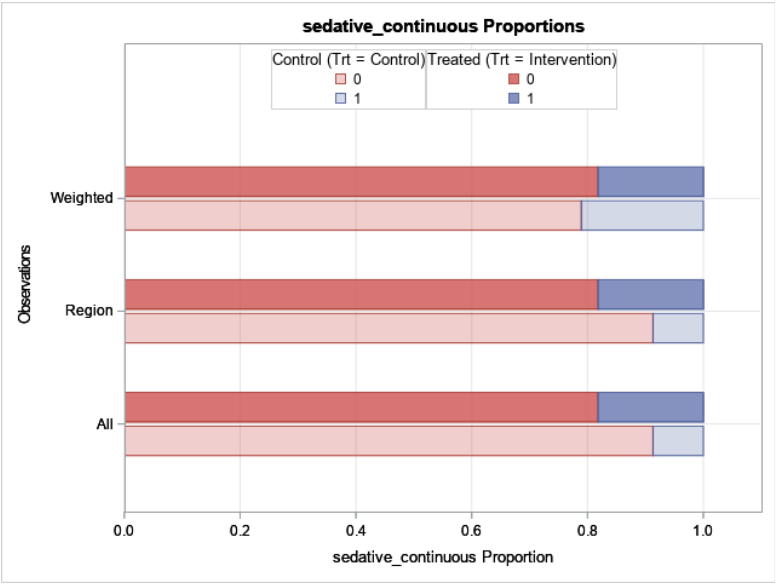




Frequencies after Weighting







Two Sample T-test without and with weighting

Primary Outcomes (unweighted)

The TTEST Procedure

Variable: fluids_first24_perkgtbsa (fluids_first24_perkgtbsa)

Trt	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Control		115	3.5486	1.5276	0.1424	0.2761	7.7588
Intervention		55	4.2387	1.6974	0.2289	1.6706	9.5392
Diff (1-2)	Pooled		-0.6901	1.5841	0.2597		
Diff (1-2)	Satterthwaite		-0.6901		0.2696		

Trt	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Control		3.5486	3.2664	3.8308	1.5276	1.3524	1.7553
Intervention		4.2387	3.7799	4.6976	1.6974	1.4290	2.0909
Diff (1-2)	Pooled	-0.6901	-1.2028	-0.1774	1.5841	1.4314	1.7737
Diff (1-2)	Satterthwaite	-0.6901	-1.2252	-0.1551			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	168	-2.66	0.0086
Satterthwaite	Unequal	97.04	-2.56	0.0120

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	54	114	1.23	0.3482

Primary Outcomes (weighted)

The TTEST Procedure

Variable: fluids_first24_perkgtbsa (fluids_first24_perkgtbsa)

Trt	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Control		115	3.7106	1.1232	0.1489	0.2761	7.7588
Intervention		55	4.2387	1.6974	0.2289	1.6706	9.5392
Diff (1-2)	Pooled		-0.5282	1.3350	0.2524		
Diff (1-2)	Satterthwaite		-0.5282		0.2730		

Trt	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Control		3.7106	3.4157	4.0055	1.1232	0.9944	1.2906
Intervention		4.2387	3.7799	4.6976	1.6974	1.4290	2.0909
Diff (1-2)	Pooled	-0.5282	-1.0265	-0.0299	1.3350	1.2062	1.4947
Diff (1-2)	Satterthwaite	-0.5282	-1.0698	0.0134			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	168	-2.09	0.0379
Satterthwaite	Unequal	100.81	-1.93	0.0559

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	54	114	2.28	0.0002

Weighted vs. Regression Model

Primary Outcomes (weighted)

The TTEST Procedure

Variable: fluids_first24_perkgtbsa (fluids_first24_perkgtbsa)

Trt	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Control		115	3.7106	1.1232	0.1489	0.2761	7.7588
Intervention		55	4.2387	1.6974	0.2289	1.6706	9.5392
Diff (1-2)	Pooled		-0.5282	1.3350	0.2524		
Diff (1-2)	Satterthwaite		-0.5282		0.2730		

Trt	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Control		3.7106	3.4157	4.0055	1.1232	0.9944	1.2906
Intervention		4.2387	3.7799	4.6976	1.6974	1.4290	2.0909
Diff (1-2)	Pooled	-0.5282	-1.0265	-0.0299	1.3350	1.2062	1.4947
Diff (1-2)	Satterthwaite	-0.5282	-1.0698	0.0134			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	168	-2.09	0.0379
Satterthwaite	Unequal	100.81	-1.93	0.0559

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	54	114	2.28	0.0002

Parameter	Estimate		Standard Error	t Value	Pr > |t|
Intercept	2.608609481	B	0.70409904	3.70	0.0003
Trt Intervention	0.566882032	B	0.26687885	2.12	0.0352
Trt Control	0.000000000	B	.	.	.
age	0.020339608		0.00777375	2.62	0.0097
inhalation_injury 0	-0.234235652	B	0.24921795	-0.94	0.3487
inhalation_injury 1	0.000000000	B	.	.	.
opioid_ome	0.000207162		0.00026978	0.77	0.4437
opioid_continuous 0	-0.767374302	B	0.35231590	-2.18	0.0308
opioid_continuous 1	0.000000000	B	.	.	.
sedative_continuous 0	0.608656548	B	0.44297346	1.37	0.1713
sedative_continuous 1	0.000000000	B	.	.	.

Extra SAS procedure: `proc causaltrt`

Analysis of Causal Effect

Parameter	Treatment Level	Estimate	Robust Std Err	Bootstrap Std Err	Wald 95% Confidence Limits		Bootstrap Bias Corrected 95% Confidence Limits		Z	Pr > |Z|
POM	Intervention	4.1070	0.2400	0.2585	3.6367	4.5773	3.6082	4.6000	17.12	<.0001
POM	Control	3.6431	0.1491	0.1633	3.3509	3.9352	3.3321	3.9595	24.44	<.0001
ATE		0.4639	0.2916	0.3244	-0.1077	1.0355	-0.1452	1.1107	1.59	0.1117

Primary Outcomes (weighted)

The TTEST Procedure

Variable: fluids_first24_perkgtbsa (fluids_first24_perkgtbsa)

Trt	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Control		115	3.7106	1.1232	0.1489	0.2761	7.7588
Intervention		55	4.2387	1.6974	0.2289	1.6706	9.5392
Diff (1-2)	Pooled		-0.5282	1.3350	0.2524		
Diff (1-2)	Satterthwaite		-0.5282		0.2730		

Trt	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Control		3.7106	3.4157	4.0055	1.1232	0.9944	1.2906
Intervention		4.2387	3.7799	4.6976	1.6974	1.4290	2.0909
Diff (1-2)	Pooled	-0.5282	-1.0265	-0.0299	1.3350	1.2062	1.4947
Diff (1-2)	Satterthwaite	-0.5282	-1.0698	0.0134			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	168	-2.09	0.0379
Satterthwaite	Unequal	100.81	-1.93	0.0559

Equality of Variances

Method	Num DF	Den DF	F Value	Pr > F
Folded F	54	114	2.28	0.0002

Variables related to outcome (fluids/kg/tbsa in first 24 hours)

- Age
- Inhalation Injury
- Time Burn Center
- Total Urine Output (first 24 hours)
- Opioid Given Continuously

Two Sample T-test without and with weighting

Primary Outcomes (weighted)

The TTEST Procedure

Variable: fluids_first24_perkgtbsa (fluids_first24_perkgtbsa)

Trt	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Control		115	3.7106	1.1232	0.1489	0.2761	7.7588
Intervention		55	4.2387	1.6974	0.2289	1.6706	9.5392
Diff (1-2)	Pooled		-0.5282	1.3350	0.2524		
Diff (1-2)	Satterthwaite		-0.5282		0.2730		

Trt	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Control		3.7106	3.4157	4.0055	1.1232	0.9944	1.2906
Intervention		4.2387	3.7799	4.6976	1.6974	1.4290	2.0909
Diff (1-2)	Pooled	-0.5282	-1.0265	-0.0299	1.3350	1.2062	1.4947
Diff (1-2)	Satterthwaite	-0.5282	-1.0698	0.0134			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	168	-2.09	0.0379
Satterthwaite	Unequal	100.81	-1.93	0.0559

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	54	114	2.28	0.0002

Primary Outcomes (New weighted)

The TTEST Procedure

Variable: fluids_first24_perkgtbsa (fluids_first24_perkgtbsa)

Trt	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Control		115	3.7455	1.0978	0.1471	0.2761	7.7588
Intervention		55	4.2387	1.6974	0.2289	1.6706	9.5392
Diff (1-2)	Pooled		-0.4933	1.3206	0.2510		
Diff (1-2)	Satterthwaite		-0.4933		0.2720		

Trt	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Control		3.7455	3.4541	4.0368	1.0978	0.9720	1.2615
Intervention		4.2387	3.7799	4.6976	1.6974	1.4290	2.0909
Diff (1-2)	Pooled	-0.4933	-0.9888	0.00223	1.3206	1.1932	1.4786
Diff (1-2)	Satterthwaite	-0.4933	-1.0330	0.0465			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	168	-1.97	0.0510
Satterthwaite	Unequal	99.739	-1.81	0.0728

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	54	114	2.39	0.0001

How about with a Categorical Outcome?

Frequency Percent Row Pct Col Pct	Table of complications_compartment_syndro by Trt		
	Trt		
complications_compartment_syndro(complications_compartment_syndrome)	Control	Intervention	Total
0	41.7592 37.31 54.40 73.36	35 31.27 45.60 63.64	76.7592 68.58
1	15.1676 13.55 43.13 26.64	20 17.87 56.87 36.36	35.1676 31.42
Total	56.9267 50.86	55 49.14	111.927 100.00

Statistics for Table of complications_compartment_syndro by Trt

Statistic	DF	Value	Prob
Chi-Square	1	1.2264	0.2681

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Trt Intervention vs Control	2.003	0.905	4.436
age	0.980	0.955	1.007
inhalation_injury 0 vs 1	0.694	0.311	1.545
opioid_ome	1.001	1.000	1.002
opioid_continuous 0 vs 1	0.511	0.191	1.372
sedative_continuous 0 vs 1	1.218	0.352	4.209

New Study That I Just Got!

- Just received as new CTSC project where the submitter specifically brought up that they wanted a propensity score match.
- Comparing non-English speakers (n=15) to English speakers (n=227) for if patients with Lung Cancer had post-op Respiratory Complications, LOS, and if they were readmitted 30 days later.
- Wanted to match on age, sex, smoking status, cardiopulmonary disease, and extent resection

Propensity Score Matching in R

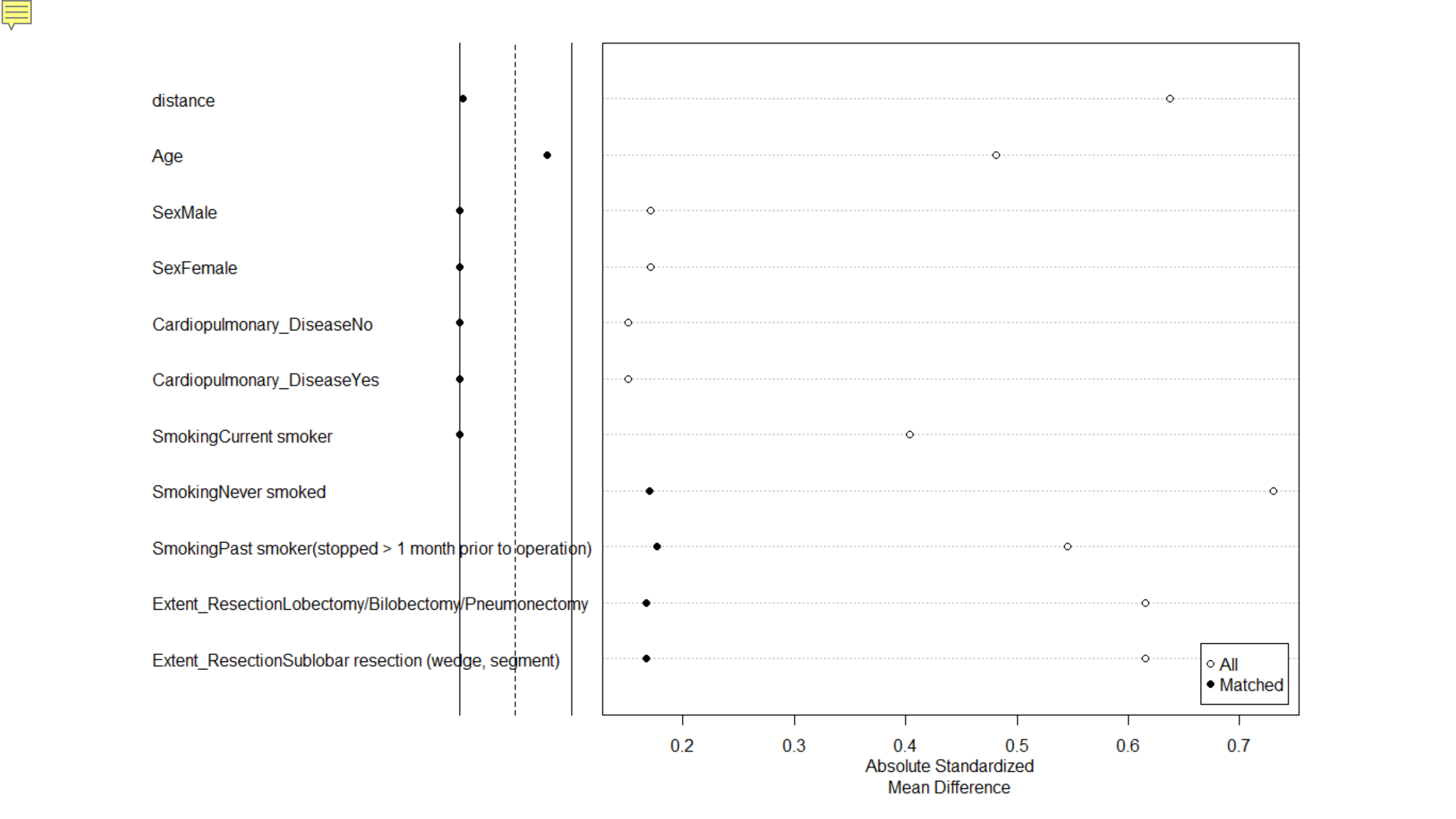
- Can do matching easily using package MatchIt.

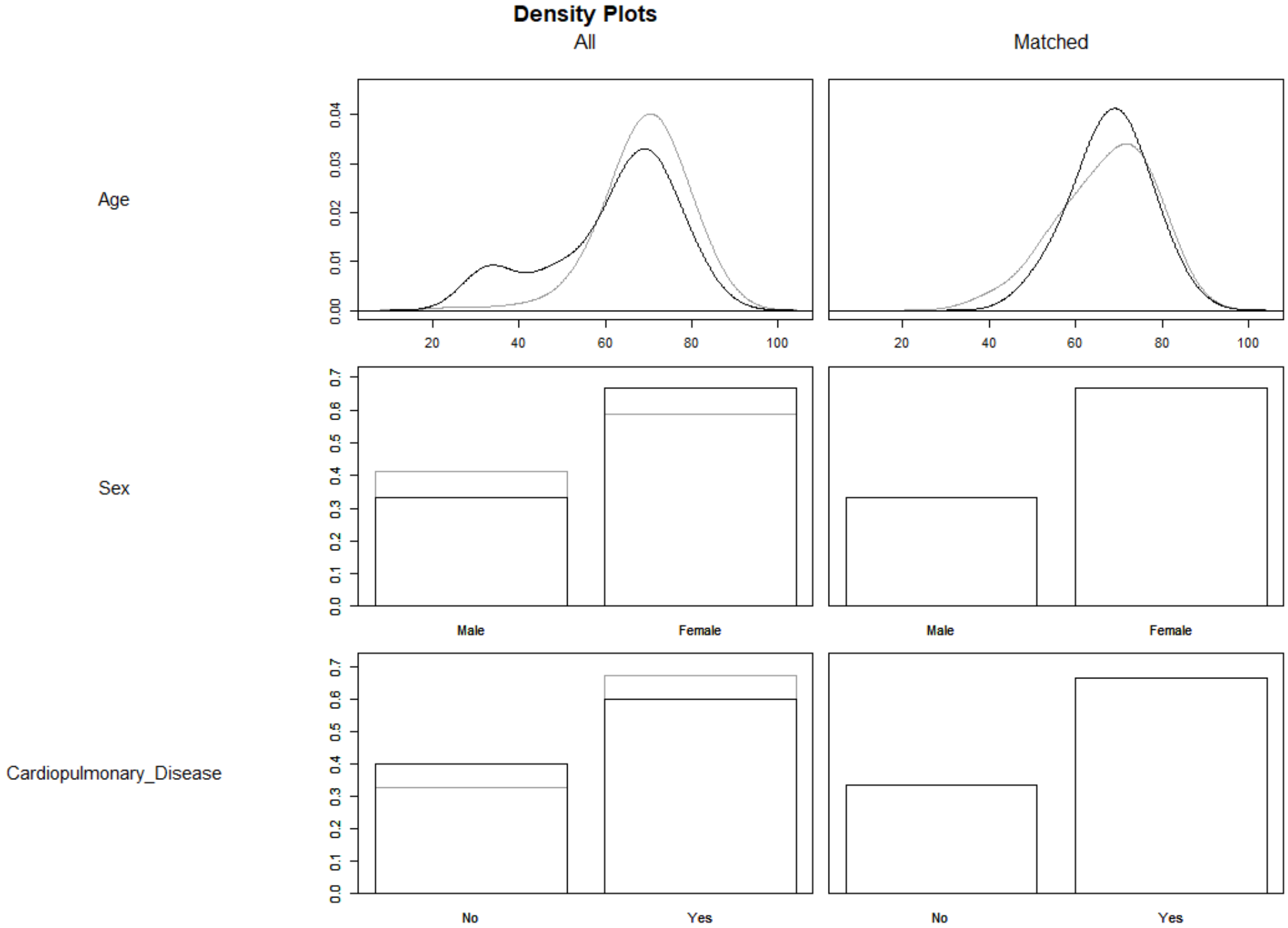
```
Call:
matchit(formula = Language ~ Age + Sex + Cardiopulmonary_Disease +
    Smoking + Extent_Resection, data = LungCancer, method = "nearest",
    distance = "glm", link = "logit", caliper = 0.2, ratio = 2)

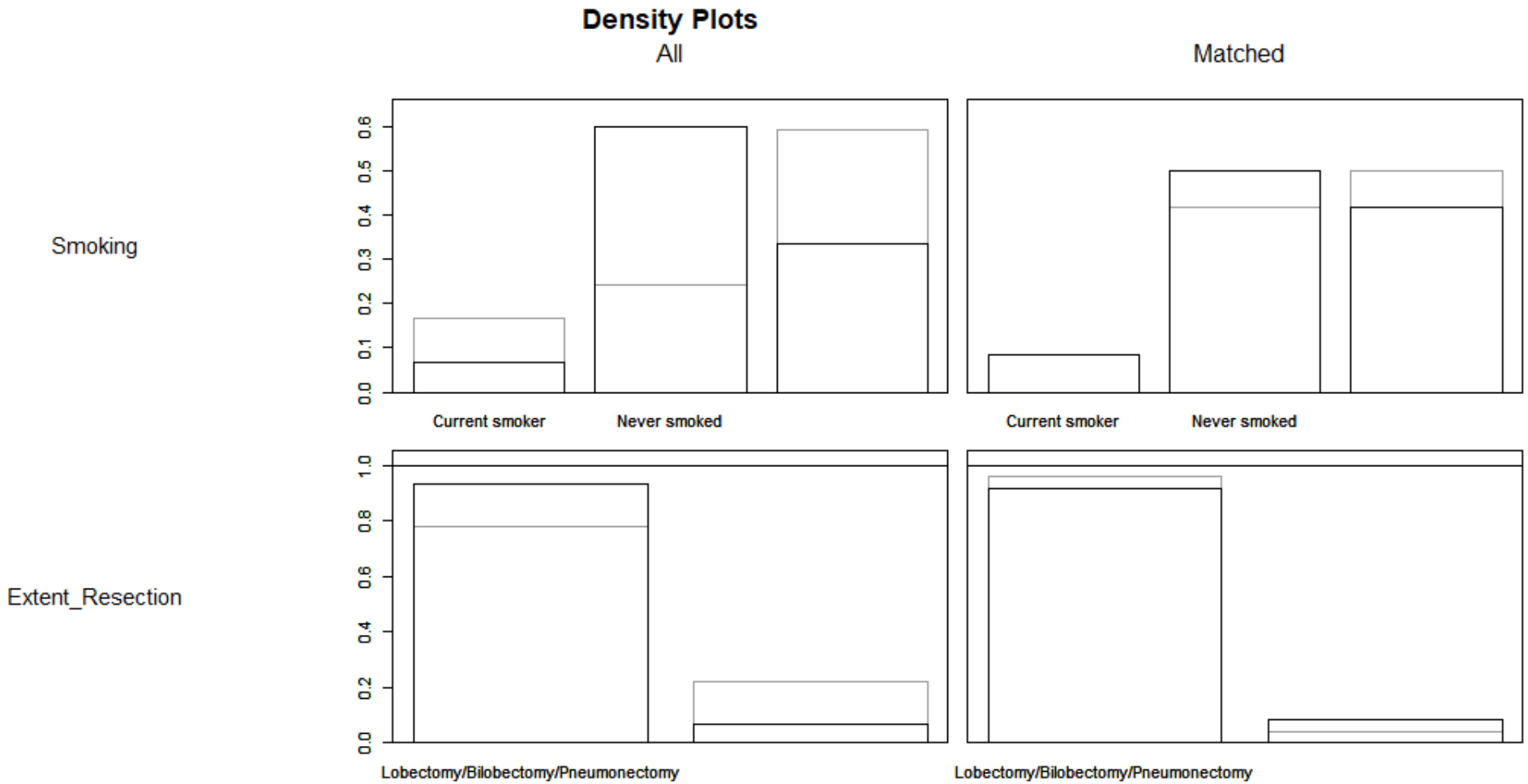
Summary of Balance for All Data:
                                                      Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean eCDF Max
distance                                                     0.1341        0.0572          0.6372     4.1204    0.2375   0.4602
Age                                                         61.9333       69.0969         -0.4812     2.5370    0.1282   0.2514
SexMale                                                      0.3333        0.4141         -0.1713          .    0.0808   0.0808
SexFemale                                                    0.6667        0.5859          0.1713          .    0.0808   0.0808
Cardiopulmonary_DiseaseNo                                    0.4000        0.3260          0.1511          .    0.0740   0.0740
Cardiopulmonary_DiseaseYes                                   0.6000        0.6740         -0.1511          .    0.0740   0.0740
SmokingCurrent smoker                                        0.0667        0.1674         -0.4038          .    0.1007   0.1007
SmokingNever smoked                                          0.6000        0.2423          0.7302          .    0.3577   0.3577
SmokingPast smoker(stopped > 1 month prior to operation)     0.3333        0.5903         -0.5451          .    0.2570   0.2570
Extent_ResectionLobectomy/Bilobectomy/Pneumonectomy          0.9333        0.7797          0.6158          .    0.1536   0.1536
Extent_ResectionSublobar resection (wedge, segment)          0.0667        0.2203         -0.6158          .    0.1536   0.1536

Summary of Balance for Matched Data:
                                                      Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean eCDF Max Std. Pair Dist.
distance                                                     0.0825        0.0822          0.0025     1.0666    0.0038   0.0833          0.0102
Age                                                         68.0833       66.9167          0.0784     0.5942    0.0463   0.1667          0.3359
SexMale                                                      0.3333        0.3333          0.0000          .    0.0000   0.0000          0.2500
SexFemale                                                    0.6667        0.6667          0.0000          .    0.0000   0.0000          0.2500
Cardiopulmonary_DiseaseNo                                    0.3333        0.3333          0.0000          .    0.0000   0.0000          0.3333
Cardiopulmonary_DiseaseYes                                   0.6667        0.6667          0.0000          .    0.0000   0.0000          0.3333
SmokingCurrent smoker                                        0.0833        0.0833          0.0000          .    0.0000   0.0000          0.1667
SmokingNever smoked                                          0.5000        0.4167          0.1701          .    0.0833   0.0833          0.3402
SmokingPast smoker(stopped > 1 month prior to operation)     0.4167        0.5000         -0.1768          .    0.0833   0.0833          0.7071
Extent_ResectionLobectomy/Bilobectomy/Pneumonectomy          0.9167        0.9583         -0.1670          .    0.0417   0.0417          0.5011
Extent_ResectionSublobar resection (wedge, segment)          0.0833        0.0417          0.1670          .    0.0417   0.0417          0.5011

Sample Sizes:
          Control Treated
All           227      15
Matched        24      12
Unmatched     203       3
Discarded       0       0
```


distance
Age
SexMale
SexFemale
CardiopulmonaryDiseaseNo
CardiopulmonaryDiseaseYes
SmokingCurrent smoker
SmokingNever smoked
SmokingPast smoker(stopped > 1 month prior to operation)
Extent_ResectionLobectomy/Bilobectomy/Pneumonectomy
Extent_ResectionSublobar resection (wedge, segment)
All
Matched
0.2
0.3
0.4
0.5
0.6
0.7
Absolute Standardized
Mean Difference


Density Plots
All
Matched
Age
Sex
Cardiopulmonary_Disease
0.00
0.01
0.02
0.03
0.04
20
40
60
80
100
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Male
Female
No
Yes


Density Plots
All
Matched
Smoking
Extent_Resection
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.8
1.0
Current smoker
Never smoked
Lobectomy/Bilobectomy/Pneumonectomy

Conclusion

- There is a lot of information on Propensity Scores but many papers had different pros and cons for when it should be used. These were the ones that came up the most often:
  - Lots of covariates to adjust for
  - The outcome is rare with common treatment

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